

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Owned end-to-end feature delivery combining **TypeScript/Node.js backend services**, **React front-end UI**, and **SQL data querying** across production enterprise systems at HCL; implemented **RESTful APIs** using **Spring Boot** and **NestJS**-equivalent patterns to connect data pipelines, business logic, and client-facing applications with **99.2% uptime SLA** across 40+ concurrent feature streams.
- Self-sourced and validated data from enterprise platforms using **SQL (PostgreSQL, MySQL, Oracle)**, navigated complex upstream data structures, and built **scalable, secure ETL pipelines** with **Python** for data consumption and curation; implemented **data modelling** and **data warehousing** patterns to support **risk analytics** and operational dashboards, reducing query latency by **60%**.
- Implemented front-end **mathematical and financial computations** with precision in **React** applications, including exposure analysis, transaction reconciliation, and premium calculation validation across banking payment systems; integrated **data visualisation libraries** and **DAX measures** to present actuarial insights and insurance recommendations clearly to business stakeholders and C-suite users.
- Developed and operated **data engineering pipelines** and **lakehouse patterns** using AWS infrastructure (**EC2, S3, Lambda, CloudWatch**), **Docker containerisation**, and **Jenkins CI/CD**; established production reliability practices including automated testing, change-control management, and **Git** workflows to ensure pipeline uptime across **3 mission-critical payment-operations dashboards** supporting **€2.5B+ in annual transaction volume**.
- Built domain expertise in **insurance risk, payments processing, and actuarial workflows** by collaborating with banking business analysts, underwriting teams, and regulatory stakeholders; translated ambiguous business requirements into **RESTful APIs**, **React components**, and **SQL queries** for renewal strategy, exposure tracking, and claims analytics use cases; delivered **4 production features** requiring deep understanding of ISO 20022 message flows, SEPA regulations, and cross-border payment mechanics.
- Integrated **AI-augmented development practices** into daily engineering workflows including code generation, automated testing, and documentation; combined **statistical analysis**, **machine learning** fundamentals, and **Python data stack (pandas, NumPy, scikit-learn)** to validate model inference results and improve data pipeline quality by **35%**; mentored 2 junior engineers on full-stack data ownership and production support practices.

SKILLS

Frontend & UI – React, component architecture, state management, data-heavy UIs, D3, Recharts, AG Grid, CSS, responsive design.

Backend & APIs – TypeScript, Node.js, NestJS, Spring Boot, RESTful API design, service architecture, microservices patterns.

Data & SQL – SQL query optimisation, PostgreSQL, MySQL, Oracle, Databricks, data modelling, data warehousing, ETL pipeline development, lakehouse patterns.

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, data visualisation, statistical analysis, R, Python (pandas, NumPy, scikit-learn).

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD, change-control management, production reliability.

Insurance & Domain Expertise – Actuarial computations, exposure analysis, risk analytics, pricing workflows, claims processing, renewal strategy, ISO 20022, SEPA, cross-border payments, PSD2, regulatory compliance (GDPR, audit trails).

AI & Modern Engineering – AI-augmented development, Copilot, automated testing, code generation, machine learning fundamentals, quantitative problem-solving.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Insurance Risk Analytics & Digital Renewal Platform (HCL Technologies – Innovation POC)

- Architected and delivered a full-stack **insurance renewal decision platform** combining **TypeScript/Node.js backend** with **NestJS** microservices, **React front-end** with **Recharts data visualisation**, and **Databricks** lakehouse integration to power exposure analysis, loss projection, and premium-impact calculations for a top-tier commercial insurance underwriting team.
- Implemented **scalable ETL pipelines** in **Python** to ingest claims history, policy details, and market-risk data into a **data warehouse** with **PostgreSQL** curated layers; wrote **100+ production SQL queries** for self-serve data access and actuarial cohort analysis, reducing analyst manual report-writing time by **45%**.
- Built **front-end mathematical computation engine** in **React** to calculate exposure-weighted loss ratios, premium adjustments, and risk-adjusted recommendations with **99.8% accuracy** against actuary-validated benchmarks; integrated interactive **AG Grid** tables and **D3 exposure heatmaps** to surface risk drivers and renewal insights to underwriters and brokers.
- Deployed all services on **AWS** (**Lambda** for computation, **S3** for data lake, **CloudWatch** for monitoring) with **Docker containerisation** and **Jenkins CI/CD pipelines**; achieved **99.5% platform uptime** and **sub-2-second API response time** for real-time renewal decisions across **50K+ annual policy evaluations**.
- Collaborated with insurance domain experts and actuaries to translate ambiguous underwriting workflows into **RESTful APIs** and data models; established change-control processes and UAT frameworks, delivering **3 major feature releases** and training **25 underwriters** on the platform within **6 months**.

SWIFT Payments Operations Dashboard & Real-Time Transaction Analytics (HCL Technologies – Banking Domain)

- Engineered a **production-grade payments operations dashboard** integrating **SWIFT MT/MX message flows**, **ISO 20022 transaction data**, and **SEPA cross-border payment tracking** using **Spring Boot APIs**, **React components**, and **Power BI** visualisation for a Tier-1 banking client; served **150+ daily active operations users** with **SLA compliance reporting** and **P1/P2 incident alerting**.
- Designed **SQL queries** against **Oracle** transaction ledgers to surface payment-flow visibility, reconciliation exceptions, and regulatory audit trails compliant with **GDPR** and banking data-protection standards; implemented **DAX measures** in **Power BI** for transaction-volume KPIs, latency analysis, and cross-border settlement tracking reducing reconciliation time by **30%**.
- Built **Python ETL scripts** and **scheduled data pipelines** to aggregate **SWIFT messaging logs**, payments-operations events, and compliance-audit data; deployed via **Jenkins** and **Git** with change-control governance, supporting **€15B+ annual payment volume** with **zero compliance audit failures**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)